



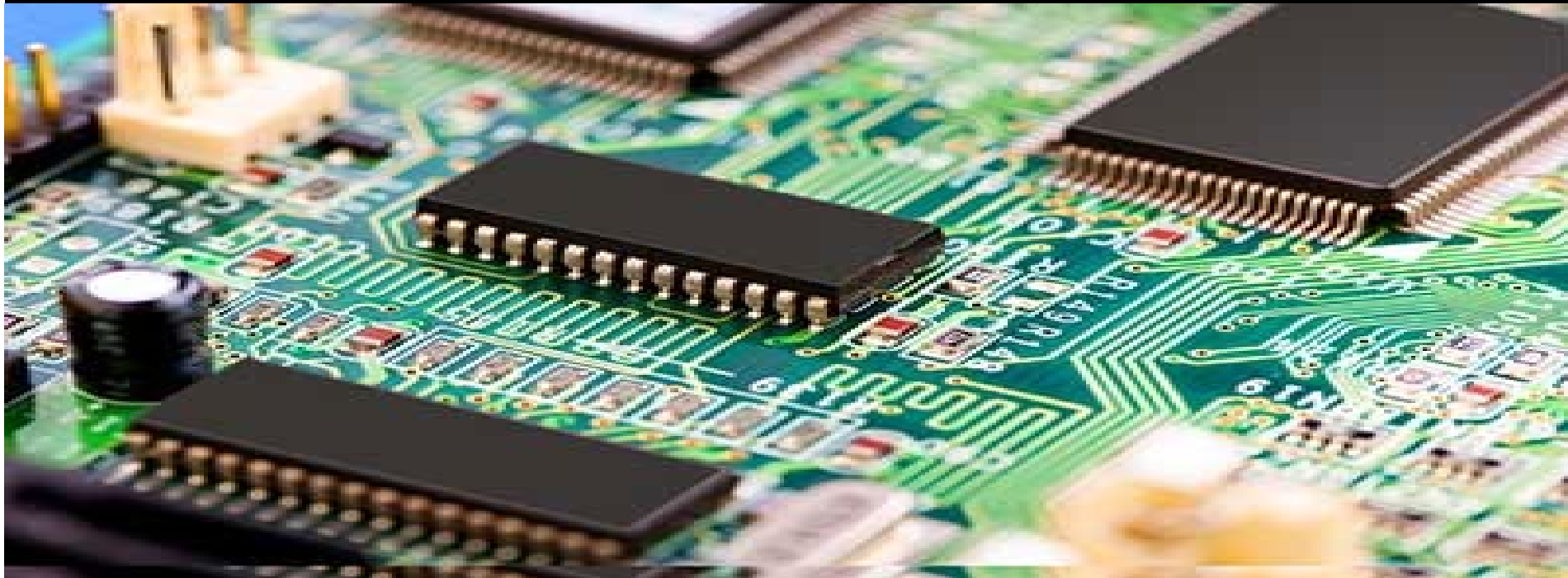
Integrated circuits



Lect 01

Introduce to fabricate ICs

Instructor /Noor .H .Jabber





Integrated circuits



A monolithic integrated circuit

commonly referred to as a "monolithic IC" or simply "IC," is a type of electronic circuit that is constructed on a single semiconductor substrate or chip.



Integrated circuits



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LM386 Low Voltage Audio Power Amplifier

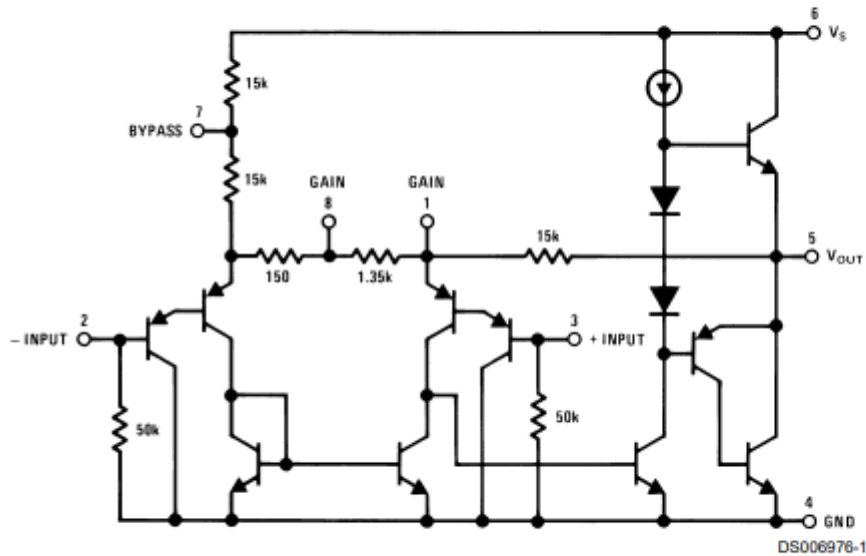


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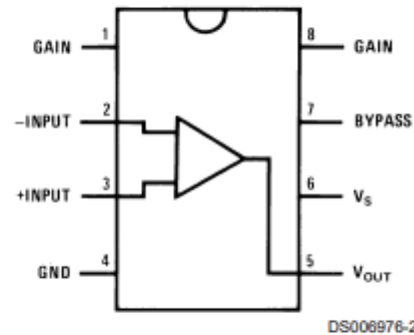


A monolithic integrated circuit

Equivalent Schematic and Connection Diagrams



Small Outline,
Molded Mini Small Outline,
and Dual-In-Line Packages



Top View
Order Number LM386M-1,
LM386MM-1, LM386N-1,
LM386N-3 or LM386N-4
See NS Package Number
M08A, MUA08A or N08E



LM386 Low Voltage Audio Power Amplifier



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A monolithic integrated circuit

advantages of monolithic integrated circuits:

- Compactness:
- Reliability:
- Cost-Efficiency
- Power Efficiency
- High Performance
- Customization:
- Reduced Size and Weight



LM386 Low Voltage Audio Power Amplifier



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Hybrid integrated circuits

advantages of monolithic integrated circuits:

- Customization
- Mixed Technology
- Reliability
- Wide Range of Applications





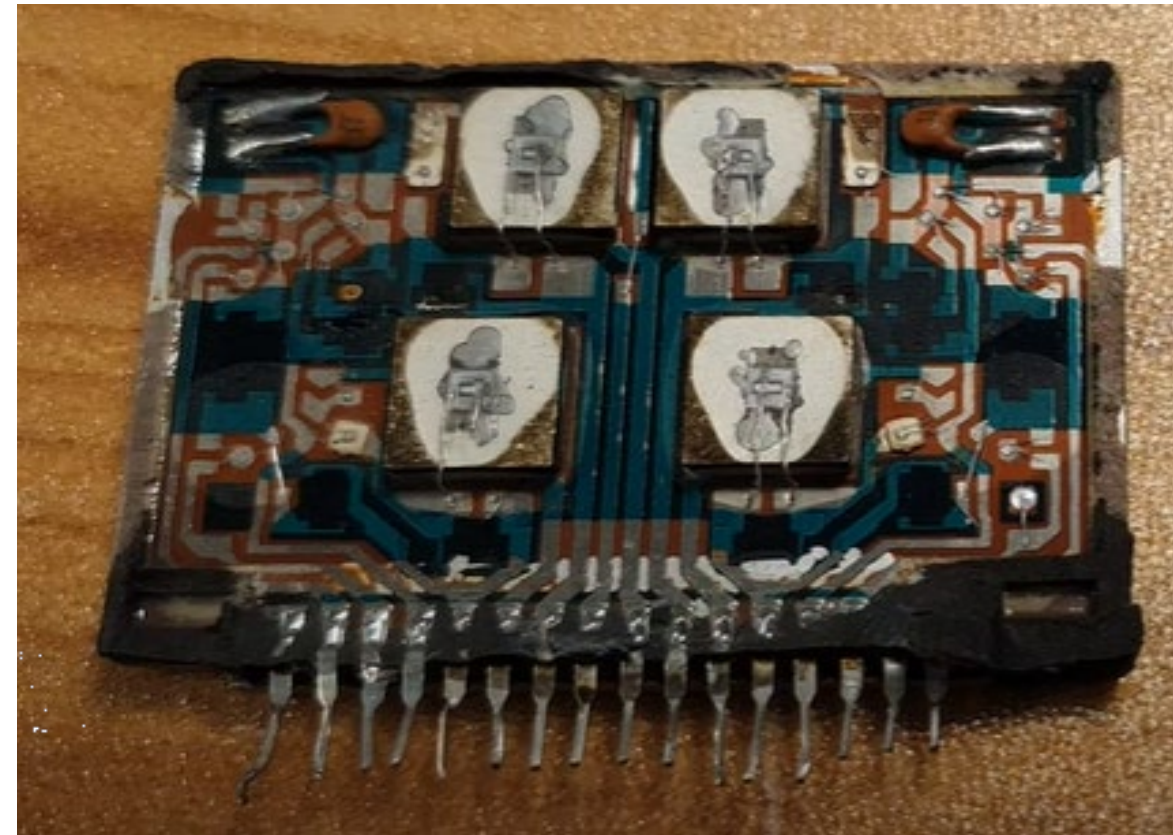
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Hybrid integrated circuits

advantages of monolithic integrated circuits:

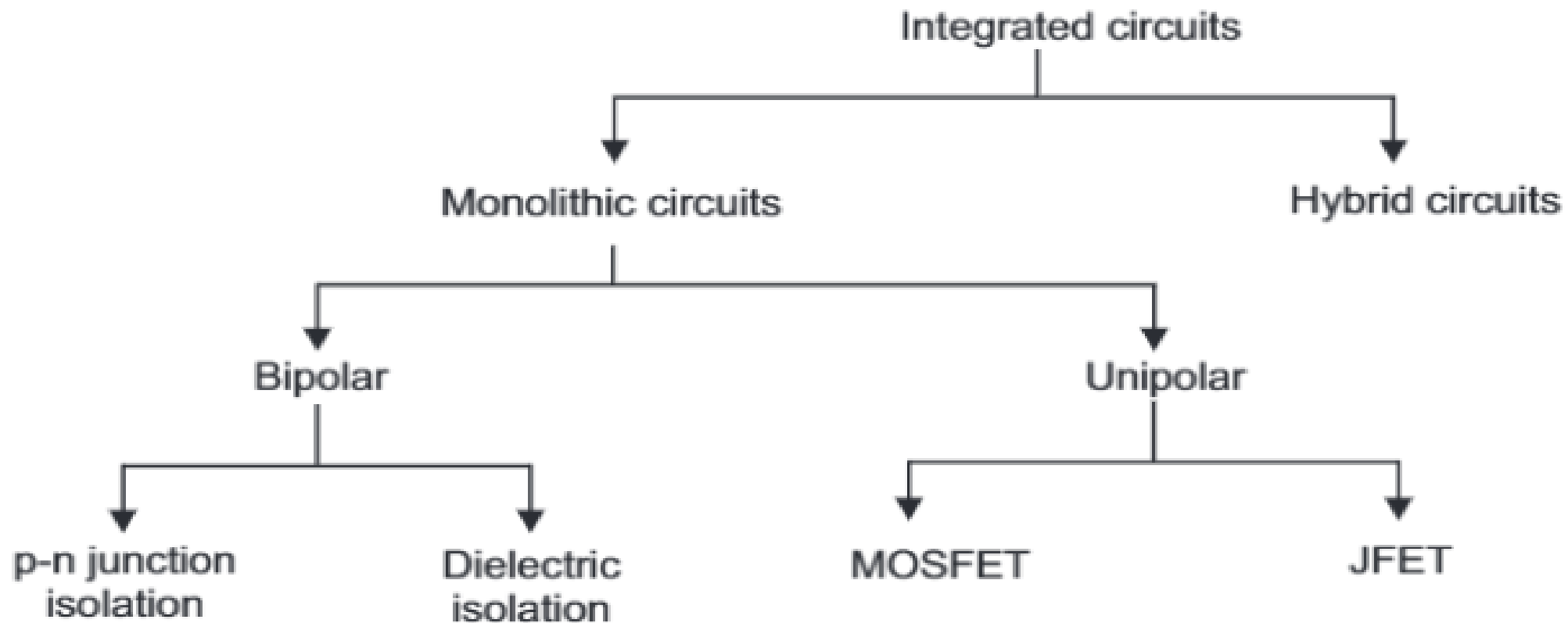
- Customization
- Mixed Technology
- Reliability
- Wide Range of Applications





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integrated circuit





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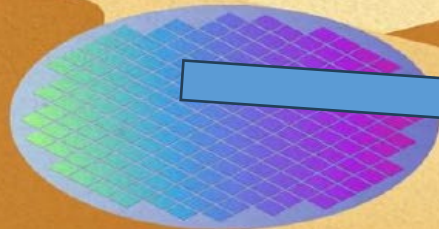
BASIC PLANAR PROCESSES

Silicon wafer

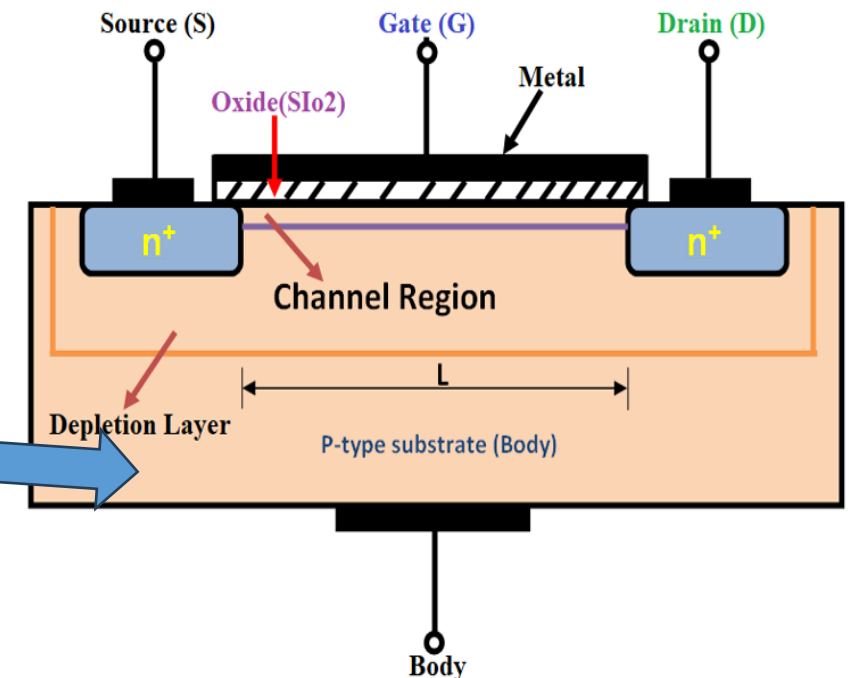
Epitaxial growth Oxidation Photolithography Diffusion Ion implantation Metallization packaging



**Silicon extraction
from sand**



How can these tiny grains of sand become a wafer?





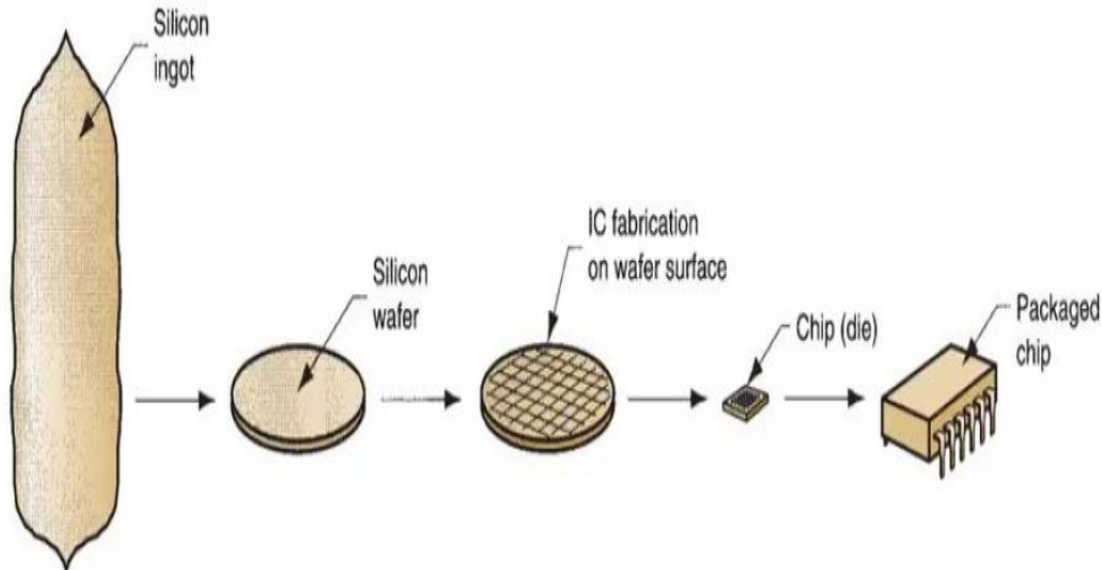
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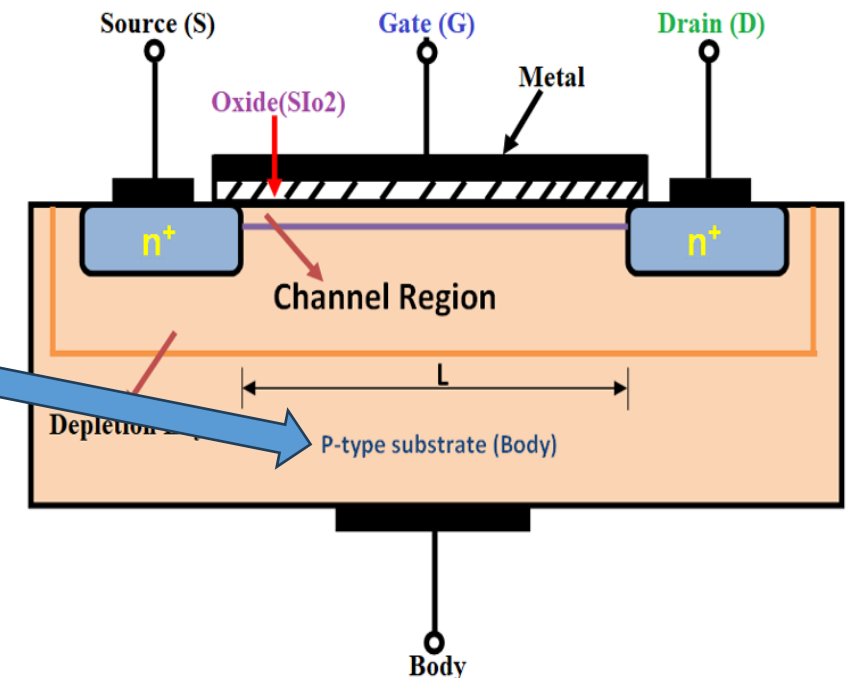
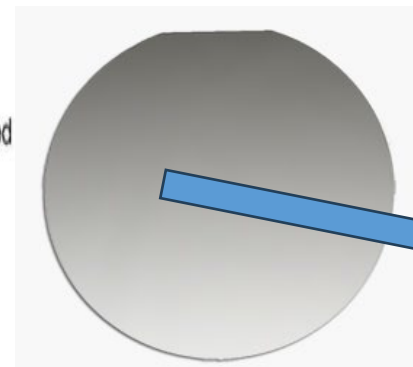
BASIC PLANAR PROCESSES

Silicon wafer

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Foundation for Semiconductor





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Silicon wafer

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Silicon Wafer Preparation:

1. **Crystal growth and doping**
2. Ingot trimming and grinding
3. Ingot slicing
4. Wafer polishing and etching
5. Wafer cleaning





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Silicon wafer

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Silicon Wafer Preparation:

1. **Crystal growth and doping:** Czochralski crystal growth process

Czochralski crystal growth process is the most often used for producing single crystal silicon ingots. The polycrystalline silicon together with an appropriate amount of dopant is put in a quartz crucible and is then placed in a furnace

2. the material is then heated to a temperature in excess of the silicon melting point of 1420°C

3. small single crystal rod of silicon called a seed crystal is then dipped into the silicon-melt and slowly pulled out

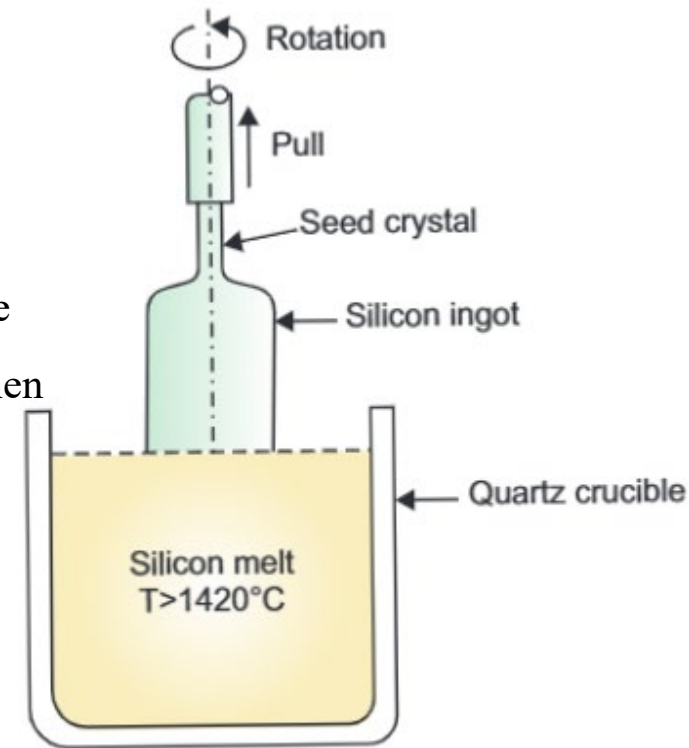


Fig. 1.5 Czochralski crystal growth



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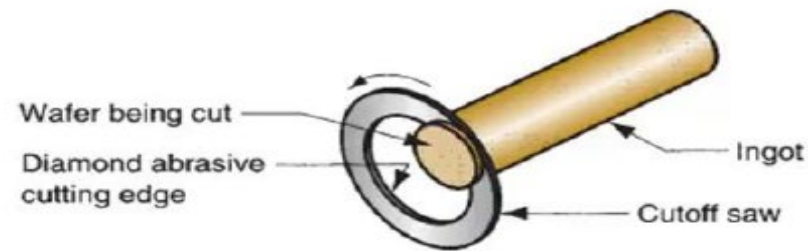
Silicon wafer

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Silicon Wafer Preparation:

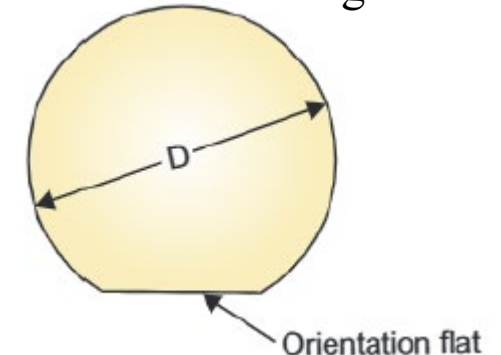
2. Ingot trimming and grinding doping:

1. The bottom portions of the ingot are cut off and the ingot surface is ground to produce an exact diameter.



2. The ingot is also ground flat slightly along the length to get a reference plane. The ingot is then sliced using a stainless steel saw blade with industrial diamonds embedded into the inner diameter cutting edge

3. wafers undergo a number of polishing steps to produce a flat surface





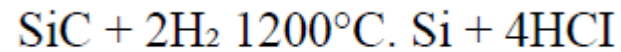
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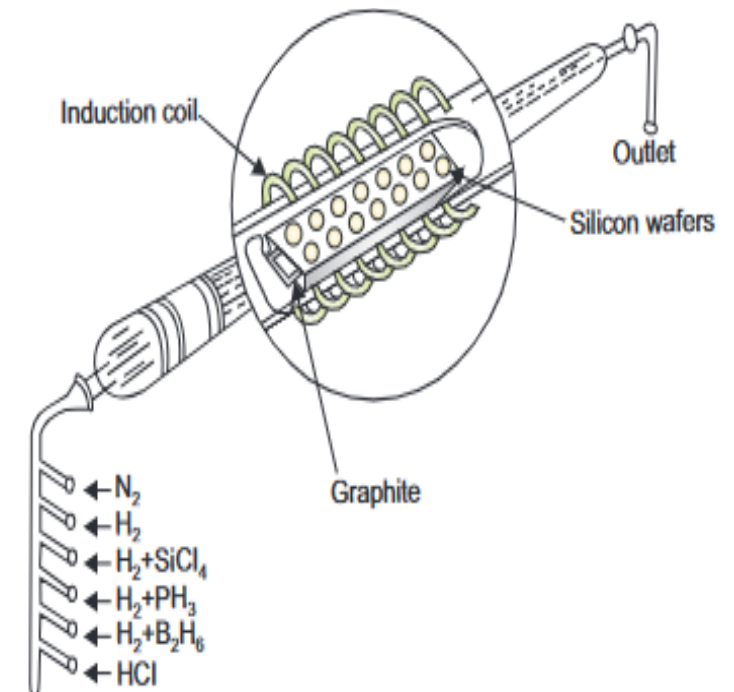
Silicon wafer Epitaxial growth Oxidation Photolithography Diffusion Ion implantation Metallization

Epitaxial growth

arranging atoms in single crystal fashion upon a single crystal substrate, so that the resulting layer is an extension of the substrate crystal structure.



silicon growth





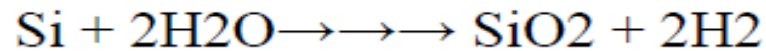
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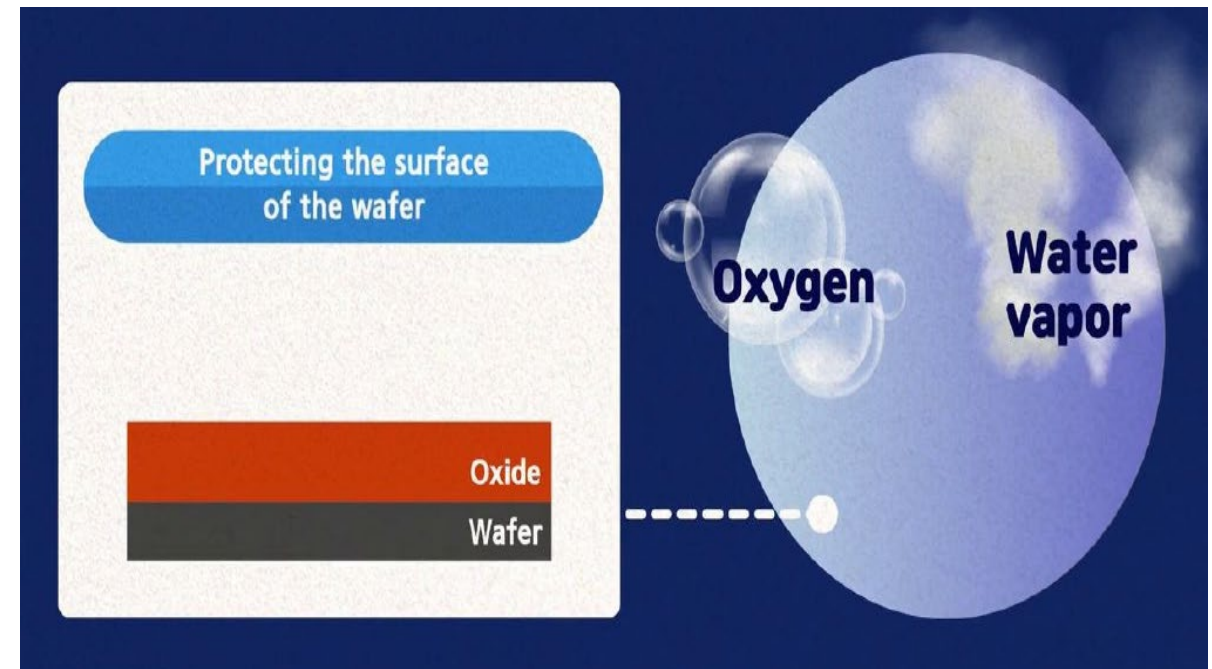
Oxidation

SiO₂ has the property of preventing the diffusion of almost all impurities through it. It serves two very important purposes



The silicon wafers are stacked up in a quartz boat and then inserted into quartz furnace tube. The Si-wafers are raised to a high temperature in the range of 950 to 1115°C and at the same time

oxidation process is called thermal oxidation because high temperature is used to grow the oxide layer

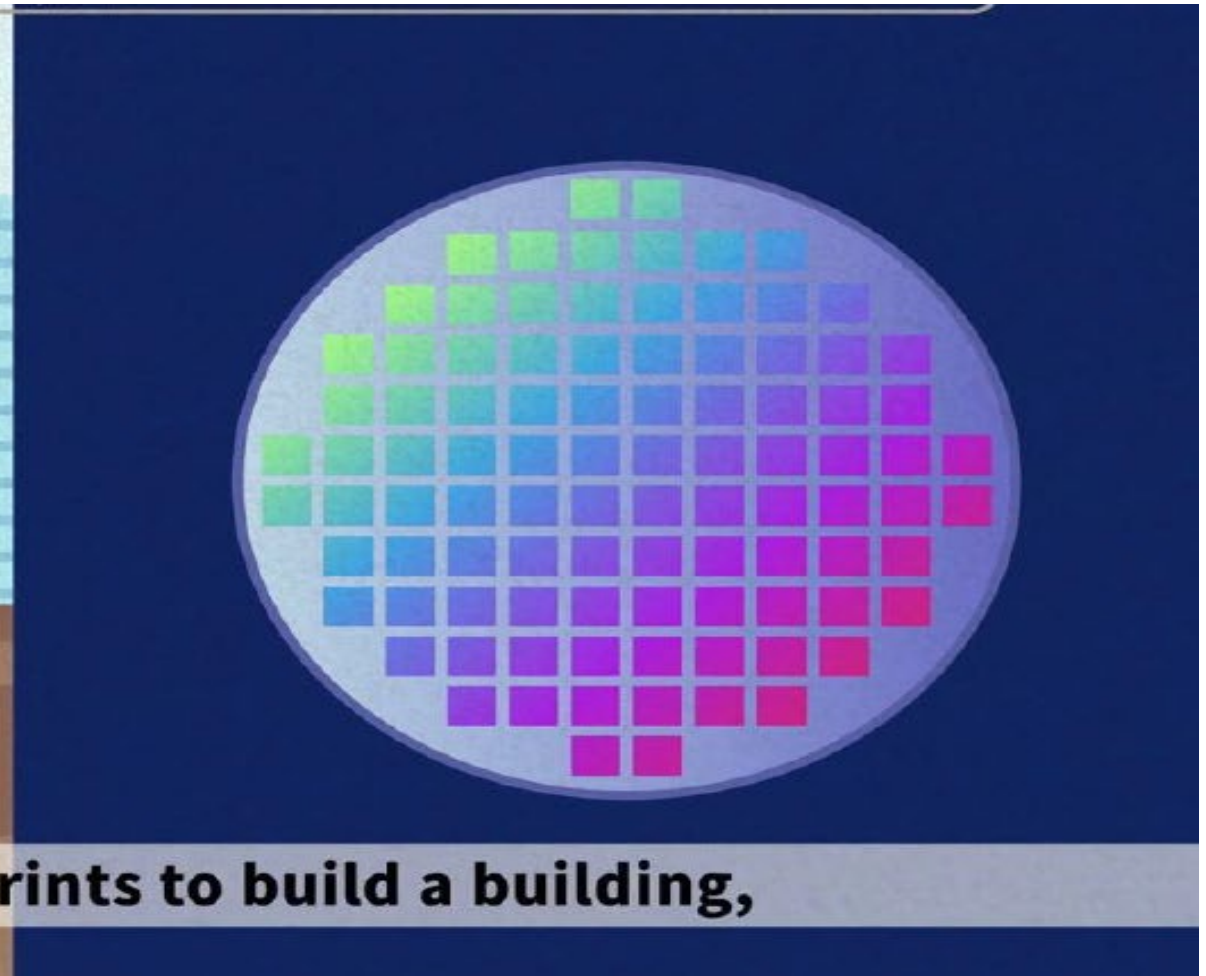
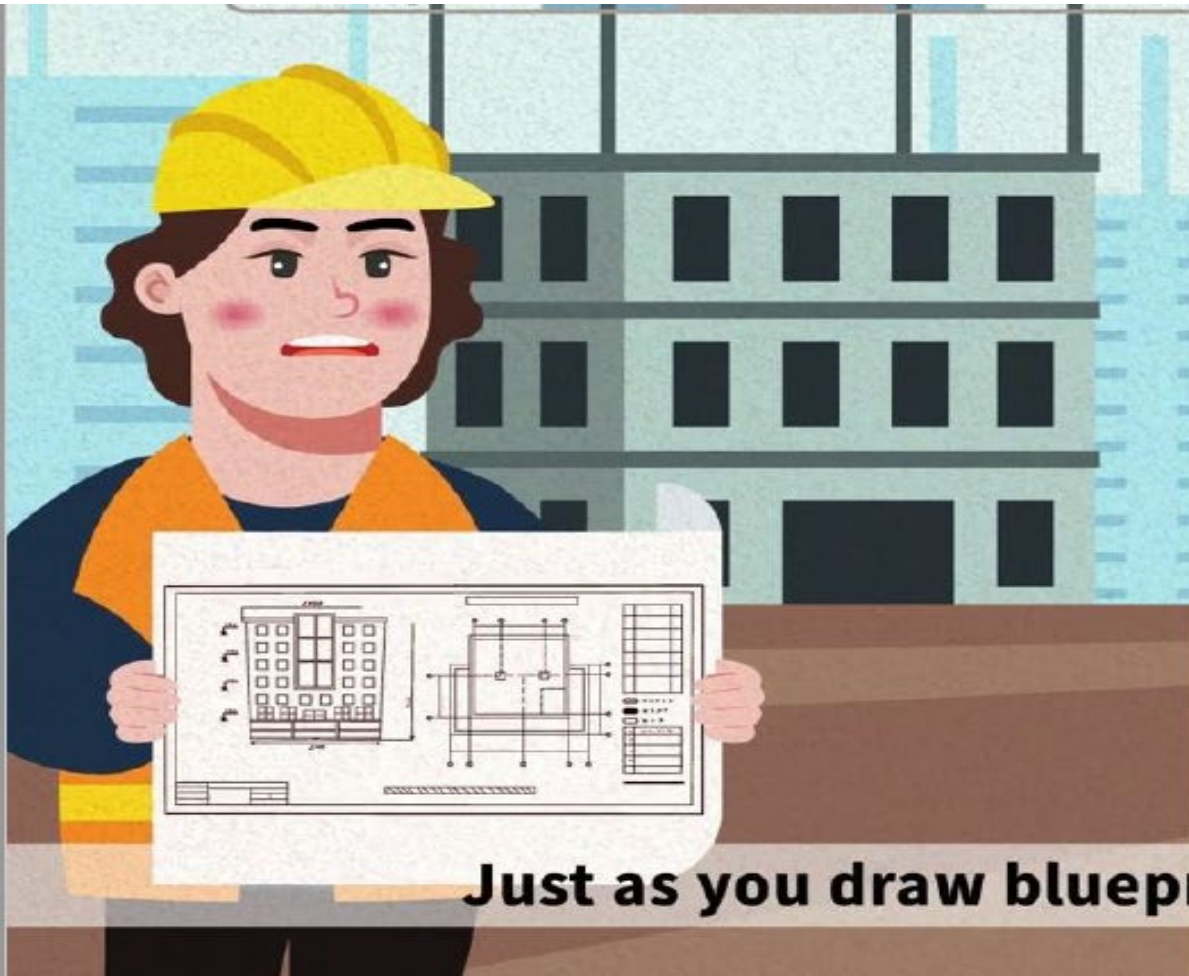




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Just as you draw blueprints to build a building,



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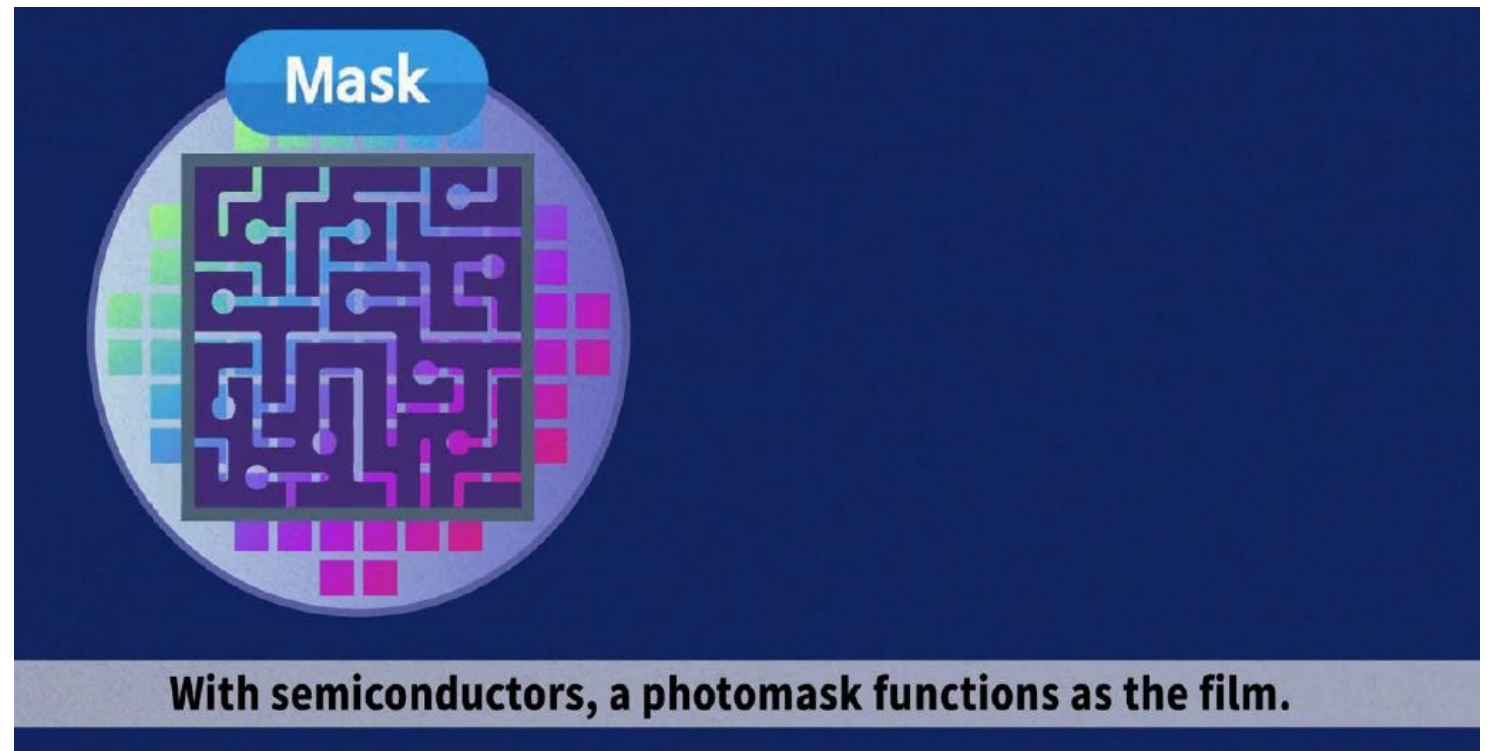
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Photolithography

to produce microscopically small circuit and device patterns on Si-wafers.

Photolithography involves two processes, namely:

- **Making of a photographic mask**
- **Photo etching**





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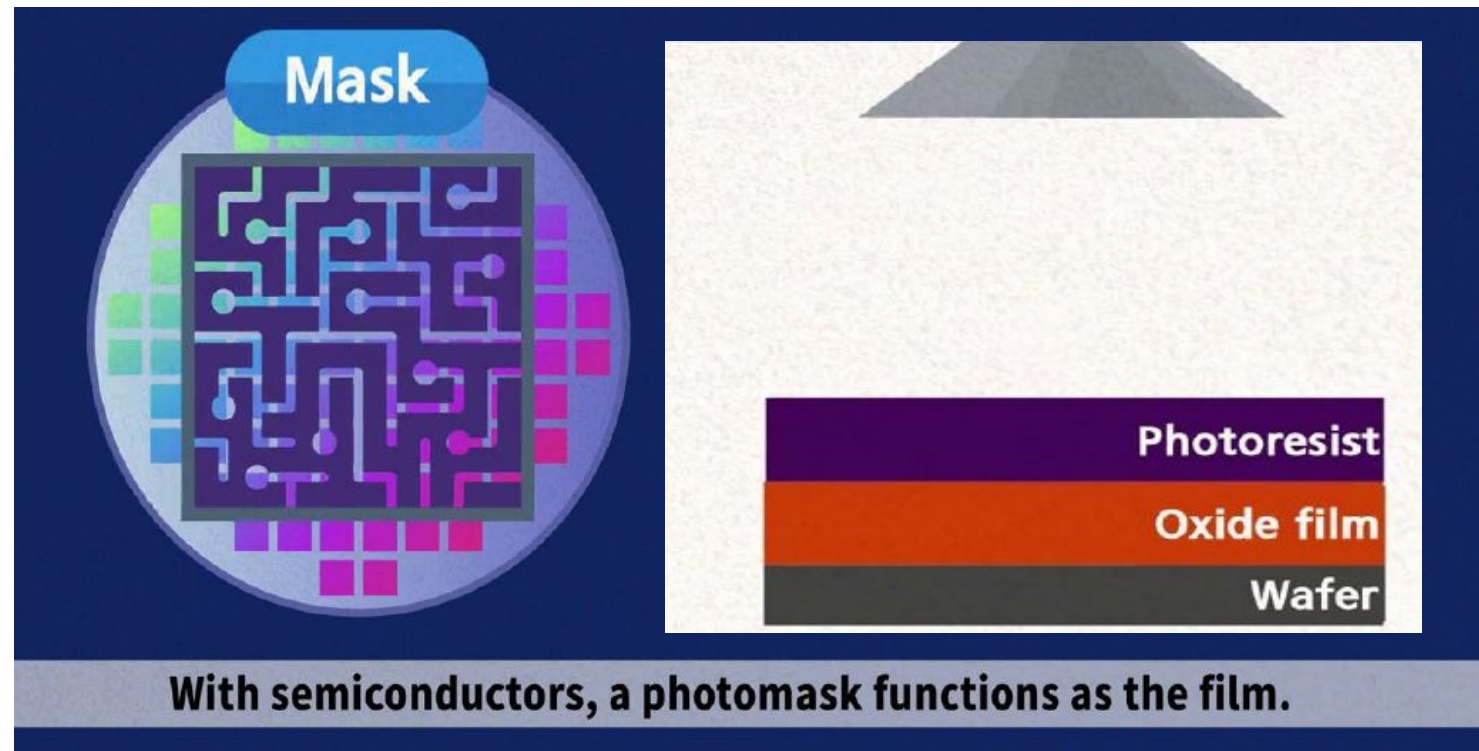
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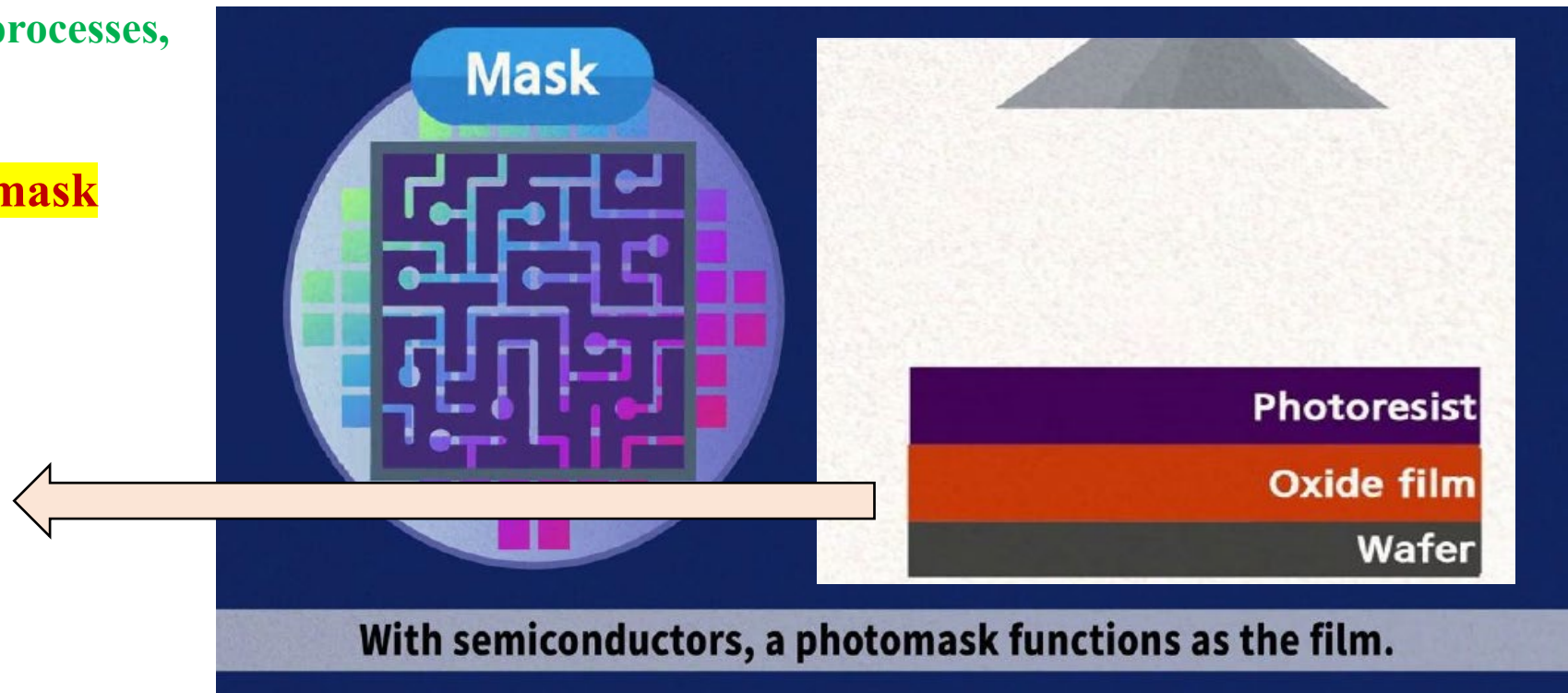
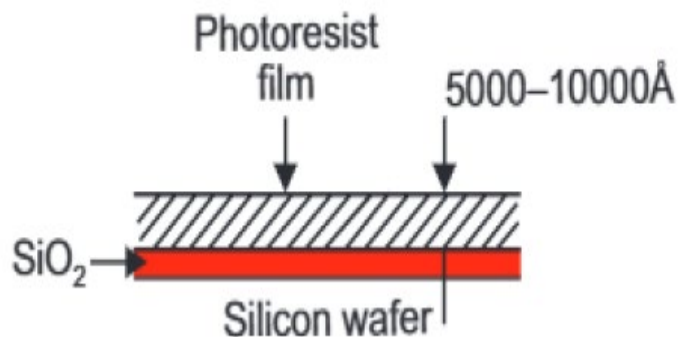
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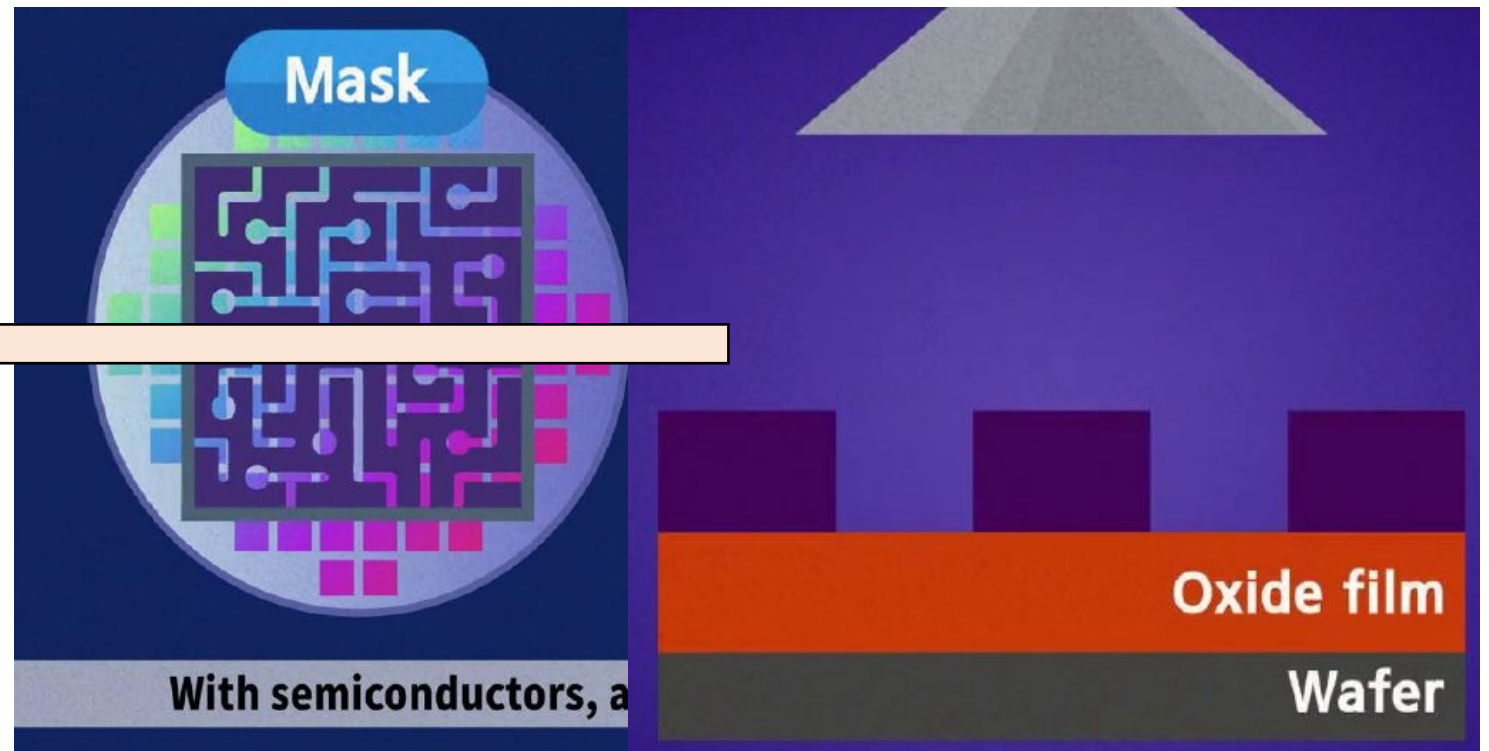
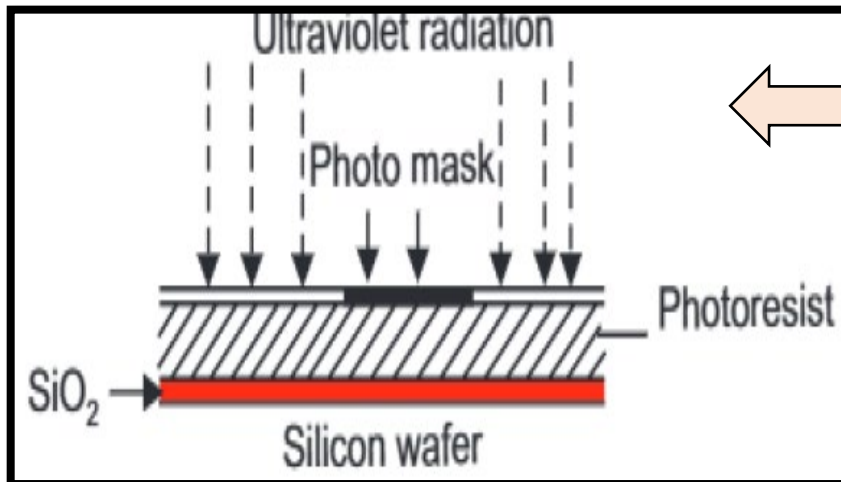
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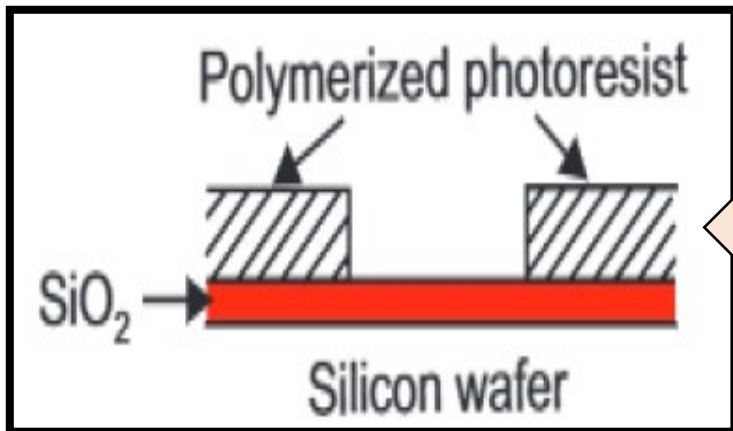
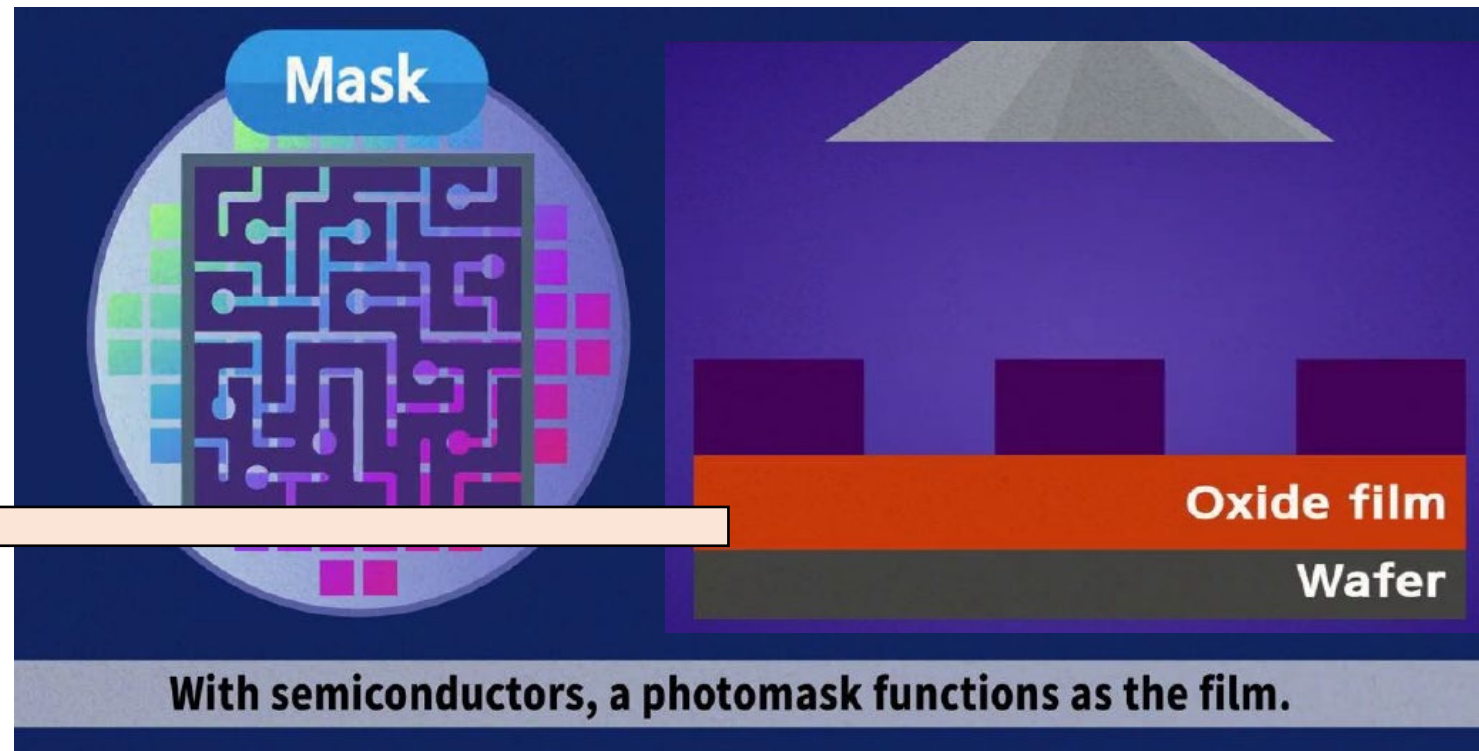
Silicon wafer Epitaxial growth Oxidation **Photolithography** Diffusion Ion implantation Metallization

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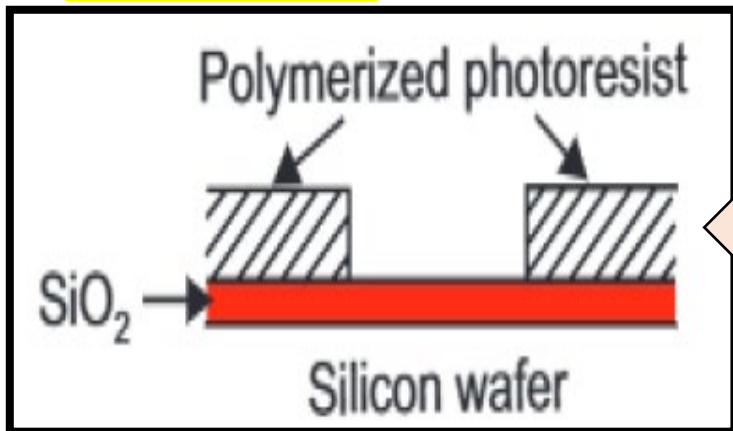
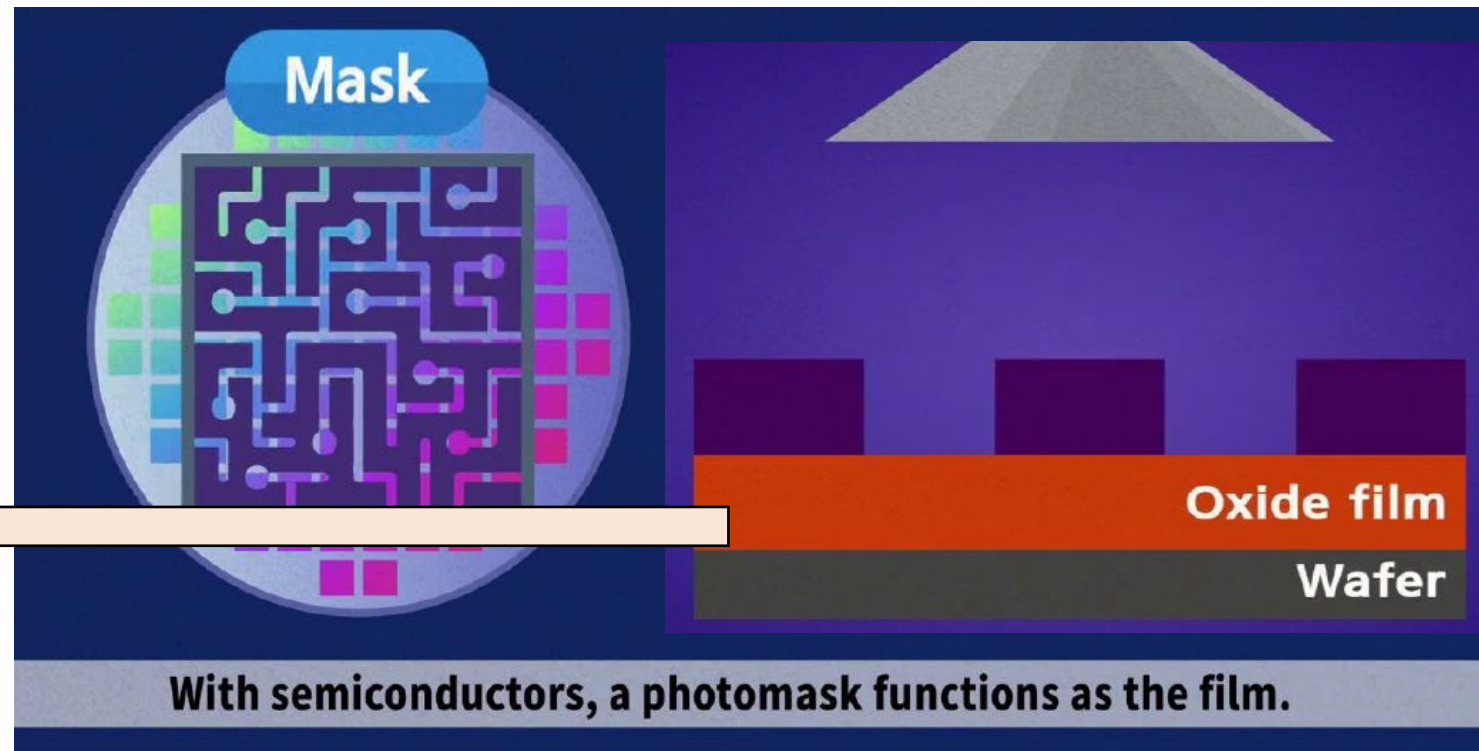
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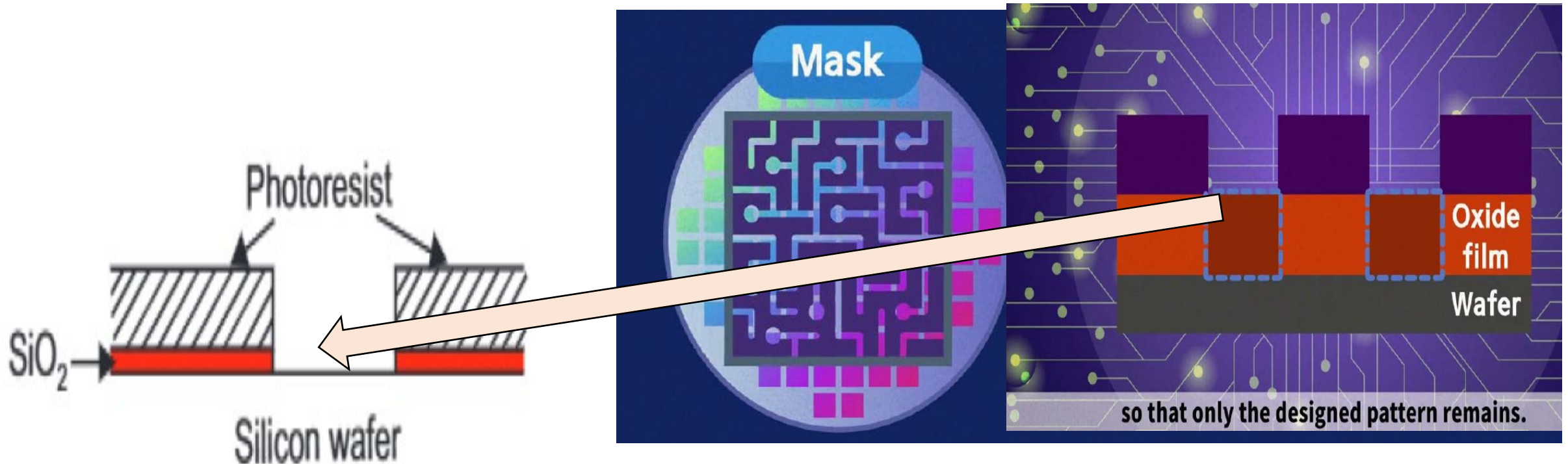
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Photo etching

- Photo-etching is used for the removal of SiO_2 from desired regions so that the desired impurities can be diffused.
- photoresist is removed with a chemical solvent (hot H_2SO_4) and mechanical abrasion. .





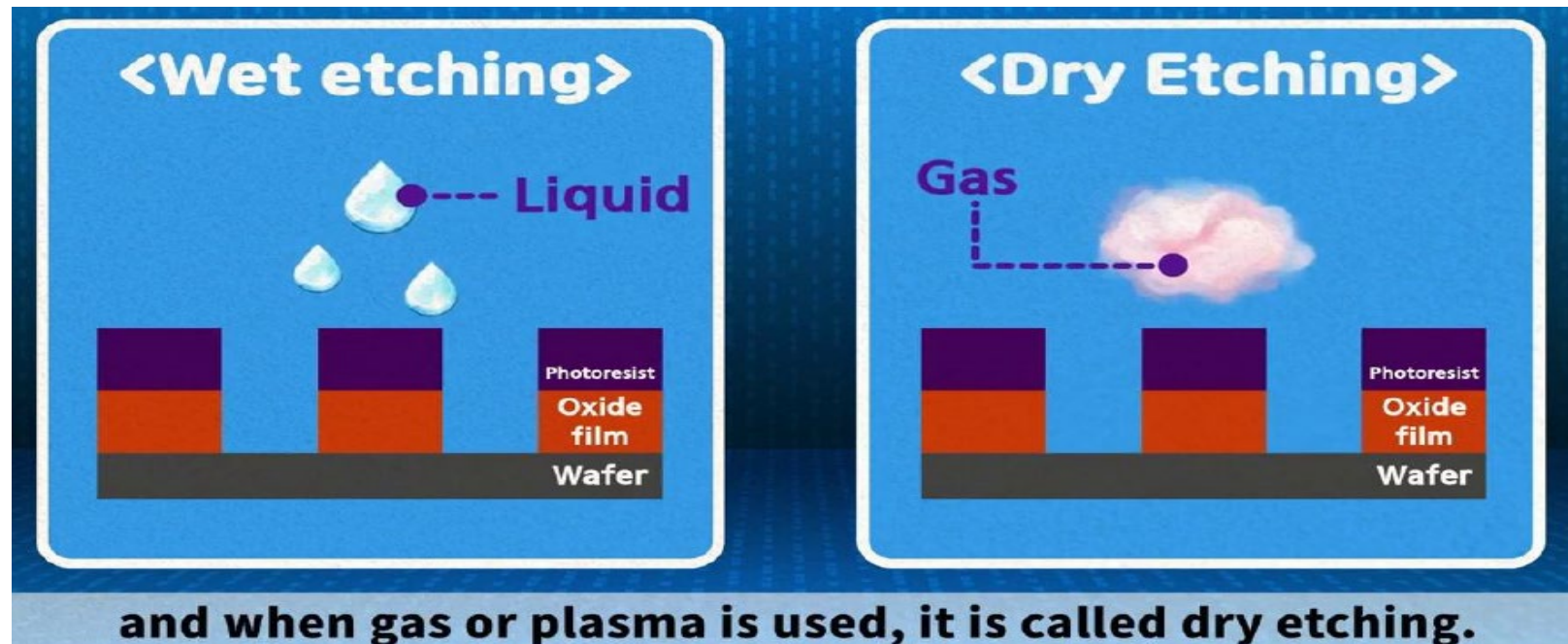
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Photo etching process

1. Wet etching
2. Dry etching (plasma etching)

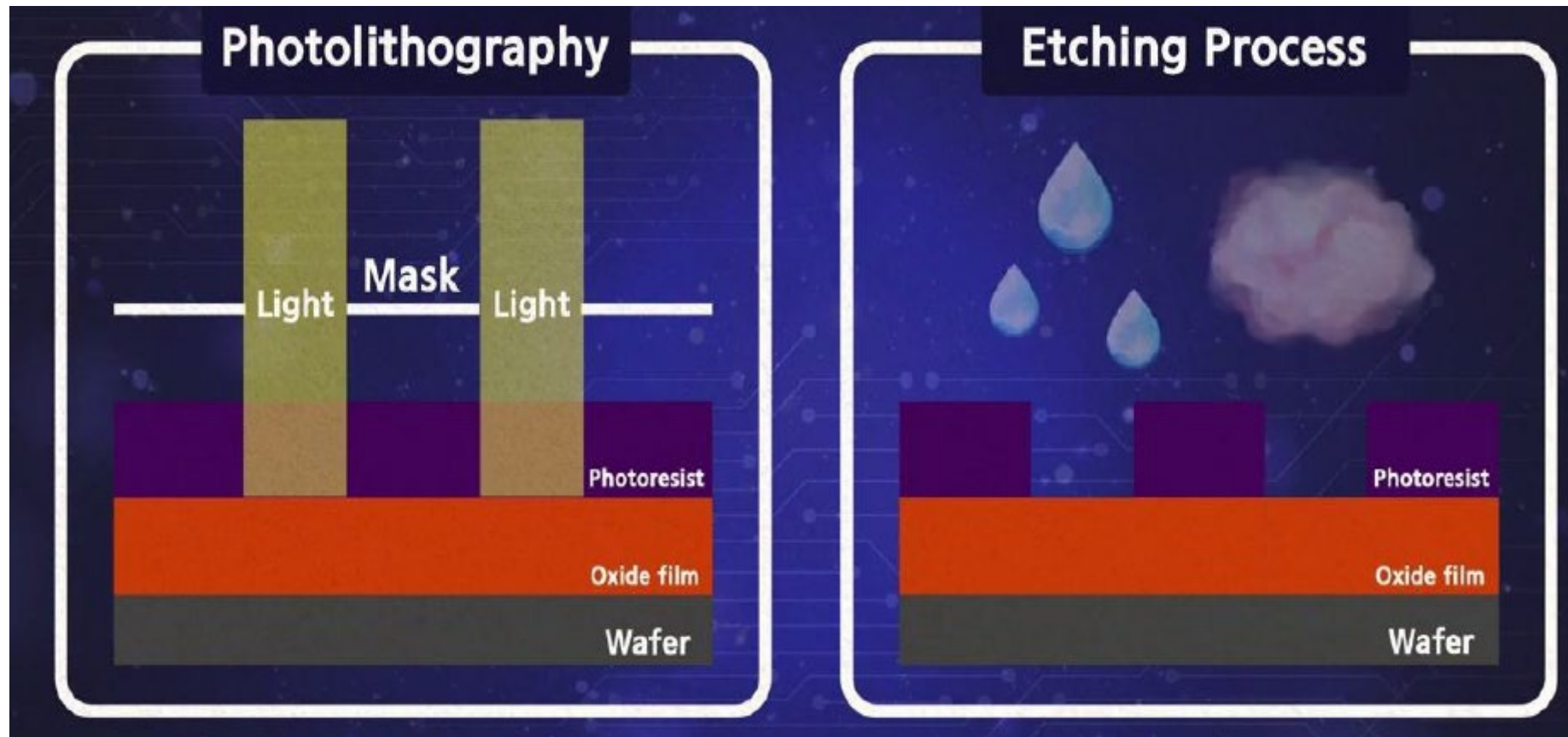




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and when gas or plasma is used, it is called dry cleaning.



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X-Ray and Electron Beam Lithography

- With conventional ultraviolet (UV) photolithography process in which the UV wavelengths used are in the **range 0.3 to 0.4 μm** ,
- mini- mum device dimensions or line widths are limited by diffraction effects to around five wavelengths or about 2 μm
- With the advent of X-ray and electron beam lithography techniques, to produce device dimensions range ($< 1 \mu\text{m}$)
- MOSFET with gate length as small as 0.25 μm have been made

Disadvantage

- very high and the exposure times very much longer than with UV photolithography only when very small device dimensions (51 μm) are needed



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Silicon wafer Epitaxial growth Oxidation Photolithography **Diffusion** Ion implantation Metallization

Another important process in the fabrication of monolithic ICs is the diffusion of impurities in the Silicon chip.

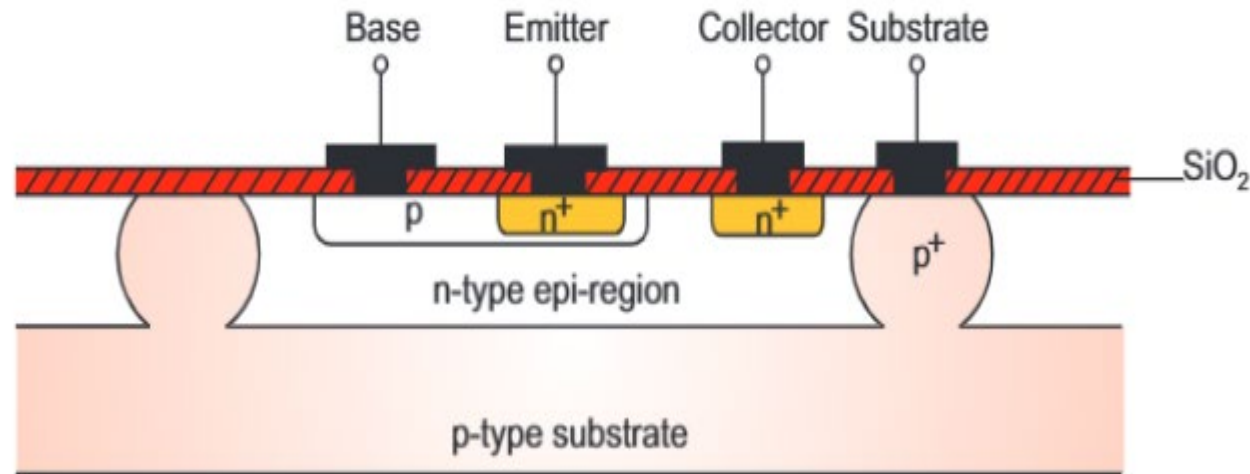


Fig. 1.9 The cross-section of an *npn* transistor showing curved junction profiles as a result of lateral diffusion



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Silicon wafer Epitaxial growth Oxidation Photolithography **Diffusion** Ion implantation Metallization

- Another important process in the fabrication of monolithic ICs is the diffusion of impurities in the Silicon chip.
- the movement of impurity atoms in a semiconductor material at high temperatures
- The depth of diffusion depends upon the time of diffusion which normally extends to 2 hours.

It is performed at low temperatures

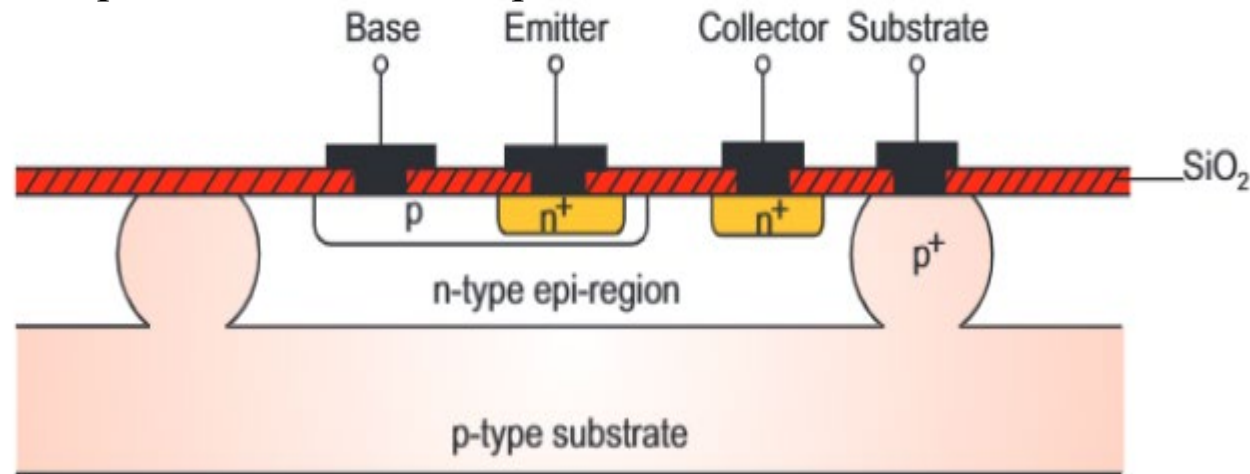


Fig. 1.9 The cross-section of an *npn* transistor showing curved junction profiles as a result of lateral diffusion

Silicon wafer Epitaxial growth Oxidation Photolithography Diffusion **Ion implantation** Metallization

- It is performed at low temperatures
- In **diffusion process**, temperature has to be controlled over a large area inside the oven
- ion implantation technique, accelerating potential and the beam current are electrically controlled from outside

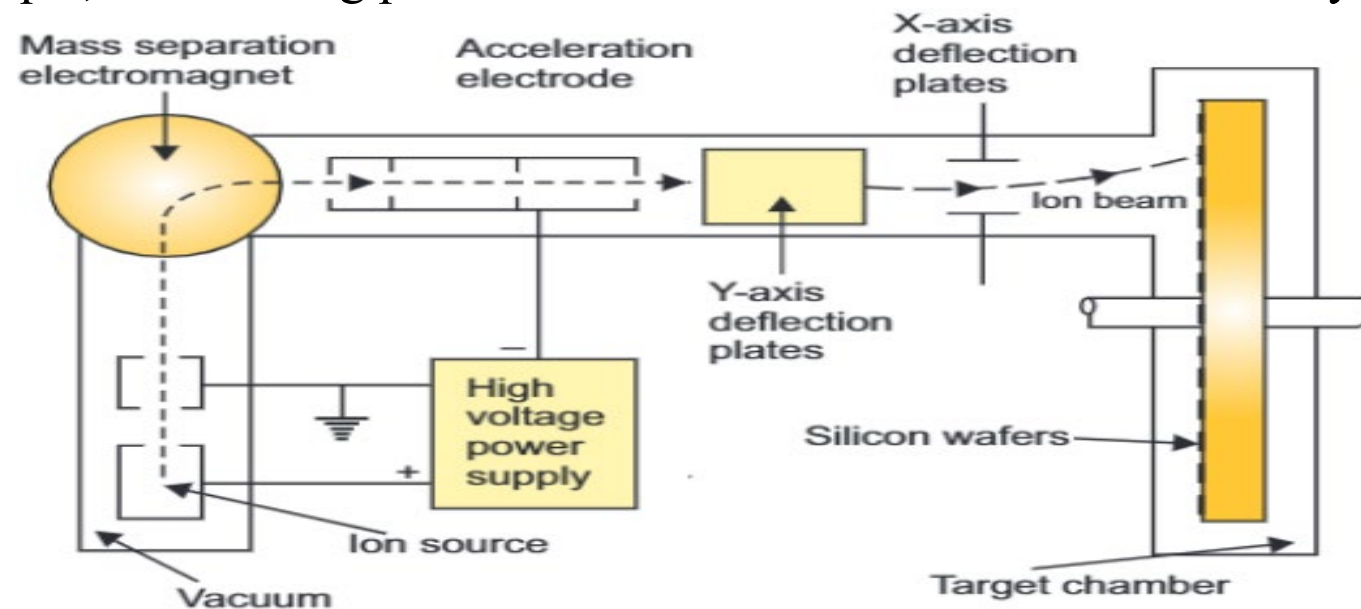


Fig. 1.10 Ion implantation system



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Isolation Techniques

- to provide electrical isolation between different components and interconnections
- **p-n Junction Isolation**
- **Dielectric Isolation**



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Isolation Techniques

- to provide electrical isolation between different components and interconnections
- **p-n Junction Isolation**
- Dielectric Isolation

P type impurities are selectively diffused into the n-type epitaxial layer so as to reach p-type substrate

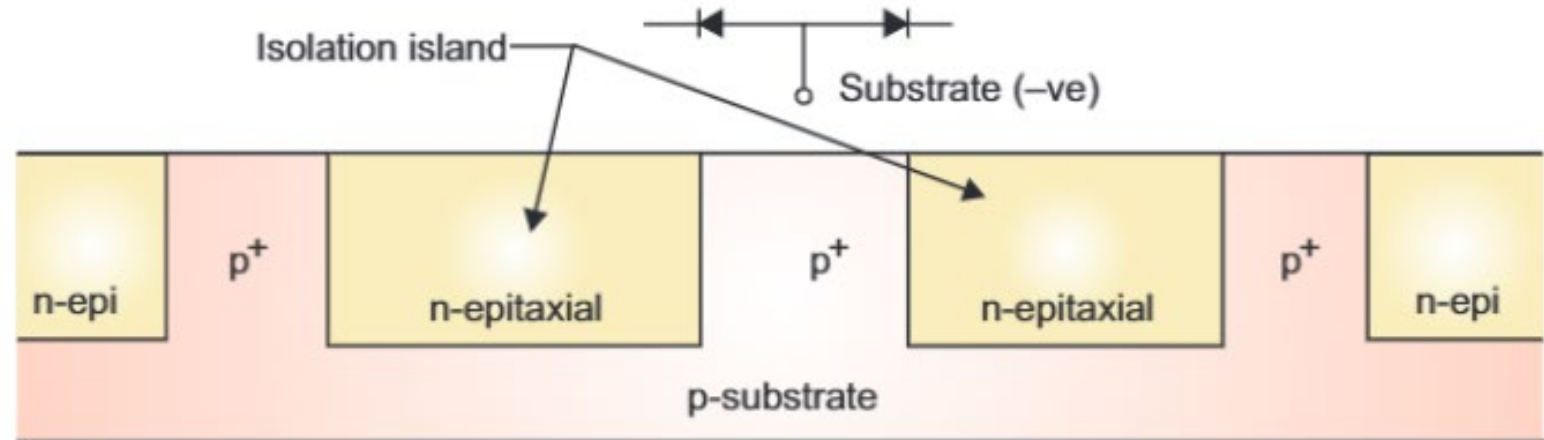


Fig. 1.11 (a) *p-n* junction isolation



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Isolation Techniques

- to provide electrical isolation between different components and interconnections

- **p-n Junction Isolation**

- **Dielectric Isolation**

- P layer of solid dielectric such as silicon dioxide or ruby completely surrounds each component,
- used for fabricating professional grade ICs required for specialized applications viz, aerospace and military,
- higher cost

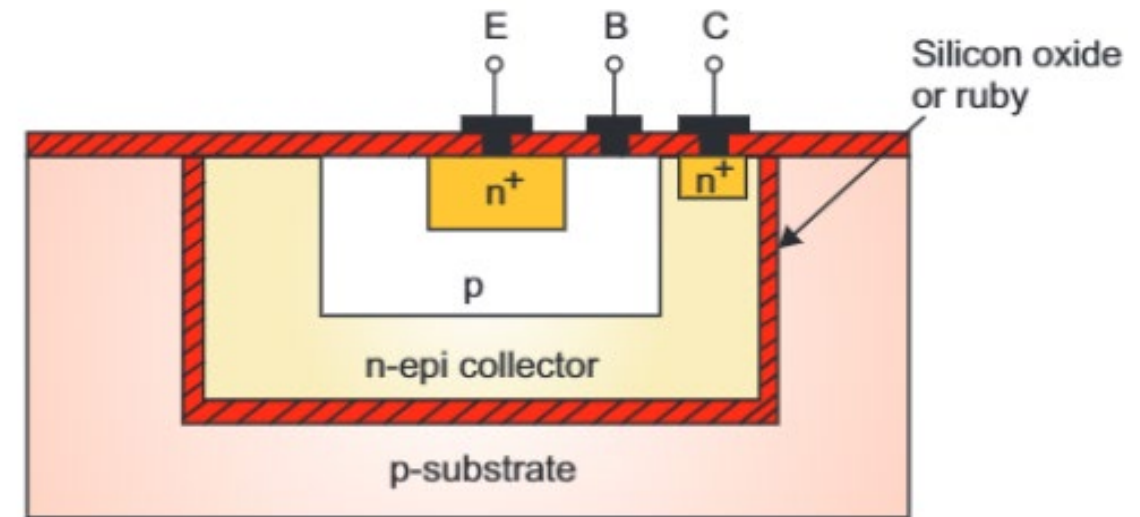


Fig. 1.11 (b) Dielectric isolation



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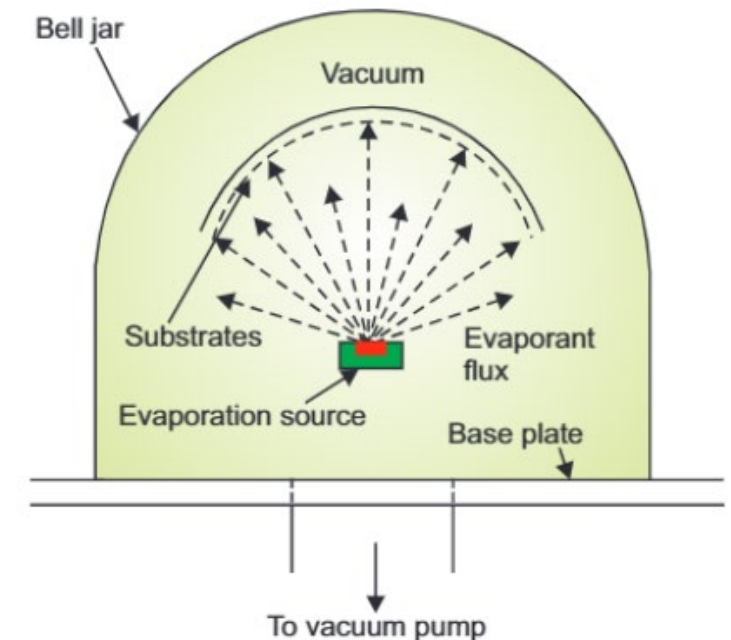
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Metallization Techniques

- The purpose of this process is to produce a thin metal film layer that will serve to make interconnections of the various components on the chip

Aluminium is usually used for the metallization of most ICs as it offers several advantages.

- It is relatively a good conductor.
- It is easy to deposit aluminium films using vacuum deposition.
- Aluminium makes good mechanical bonds with silicon.
- Aluminium forms low resistance





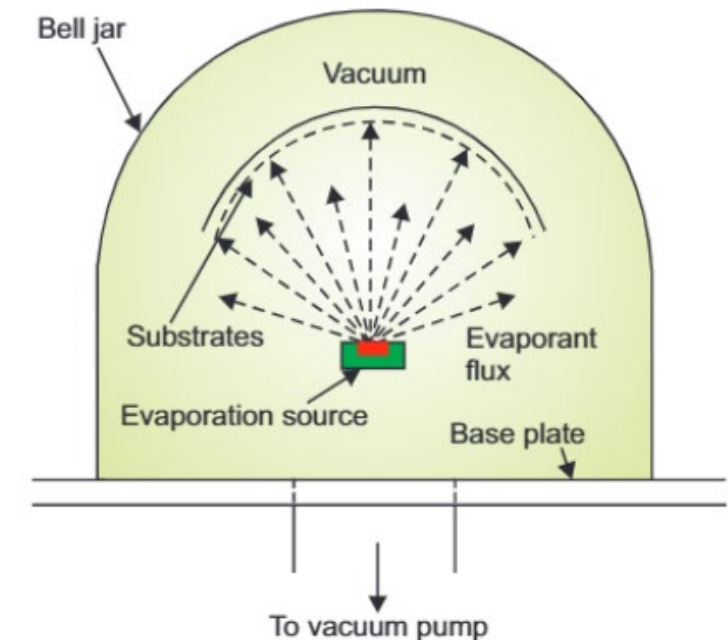
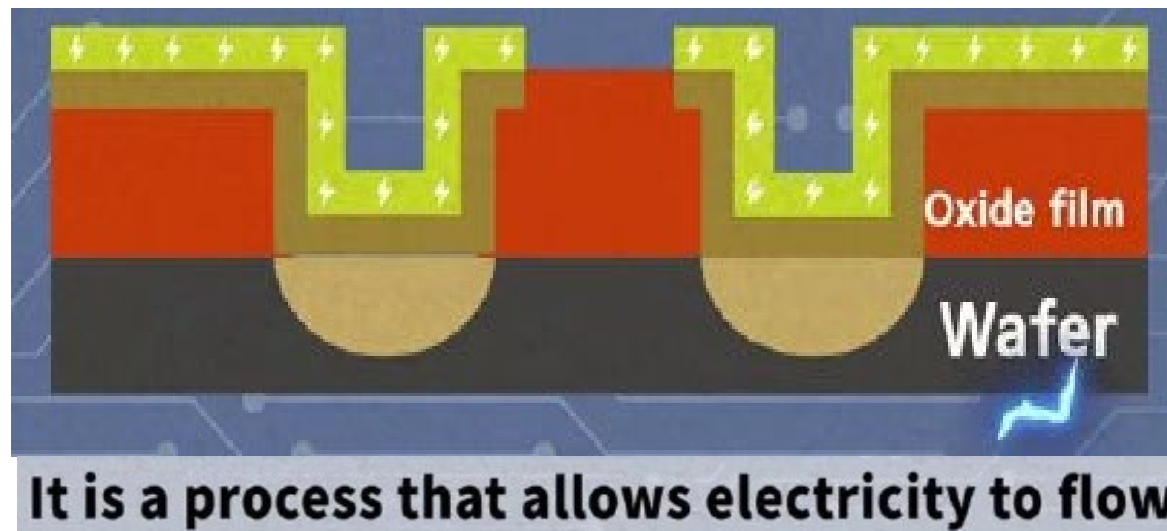
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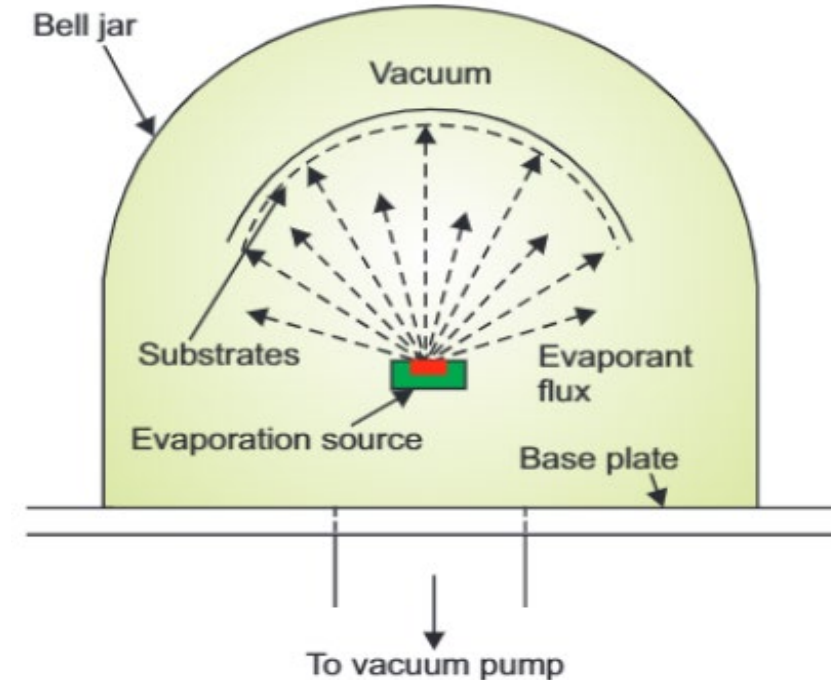
Silicon wafer Epitaxial growth Oxidation Photolithography Diffusion Ion implantation Isolation Metallization
Assembly Processing and Packaging

Assembly Processing and Packaging

scribing and cleaving used for separation makes use of a diamond tipped tool to cut lines into the surface of the wafer along the rectangular grid separating the individual chip

there are three different package configurations available.

- TO-5 glass metal package
- Ceramic flat package
- Dual-in-line (ceramic or plastic type)





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Assembly Processing and Packaging

Assembly Processing and Packaging

